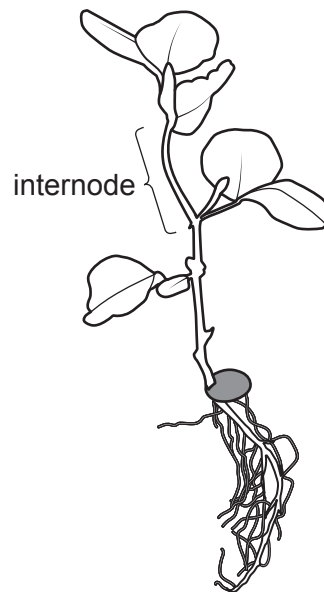


- Fig. 2.1 shows a diagram of a pea plant.



Mendel observed that the pea plants he grew either had tall stems or dwarf (short) stems. In his investigation, Mendel carried out crosses using pea plants with these two phenotypes.

- It is now known that the stem length trait in pea plants is controlled by one gene that has two alleles:

- Describe a cross that could be carried out **and** how the results of the cross could be analysed to determine the genotype of a pea plant with a tall stem.

[4]

- (b) The scientists P W Brian and H G Hemming identified that the difference in the length of the stem in pea plants was associated with the presence or absence of gibberellin. They published their findings in 1955.

- (i) Gibberellin leads to a response in plant cells by binding to specific receptor molecules.

State the term used to describe a molecule, such as gibberellin, that binds to specific receptor molecules and leads to a response in cells.

..... [1]

- (ii) Suggest the response of the cells in the internode region of the stem, as labelled in Fig. 2.1, to the presence of gibberellin **and** describe how this response affects the trait investigated by Mendel.

.....
.....
.....
.....
..... [2]

[Total: 7]